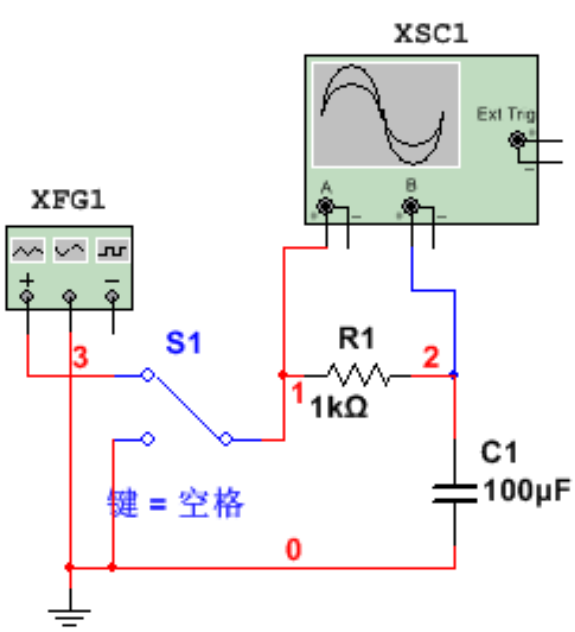
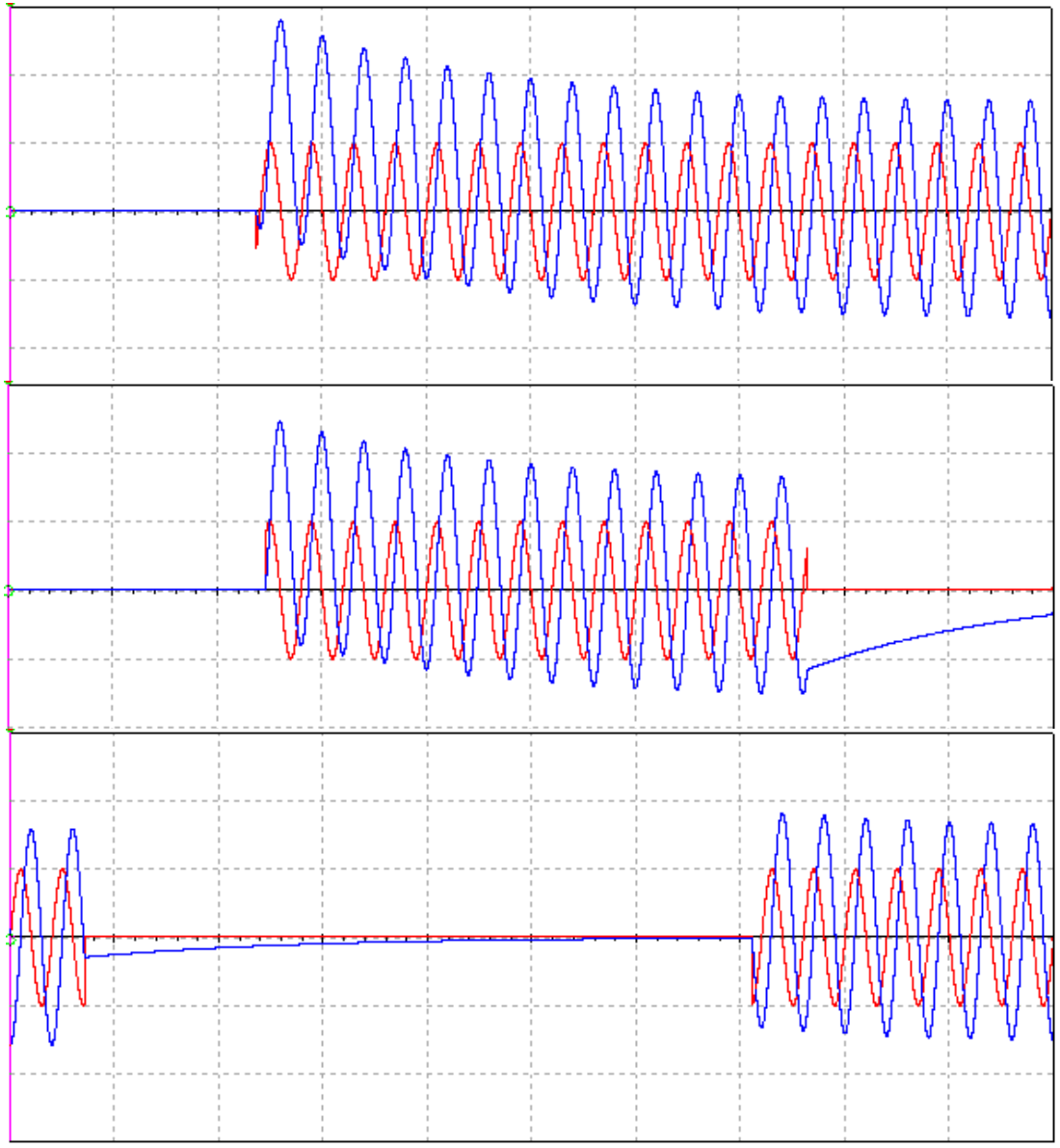
正弦波激励暂态响应示例

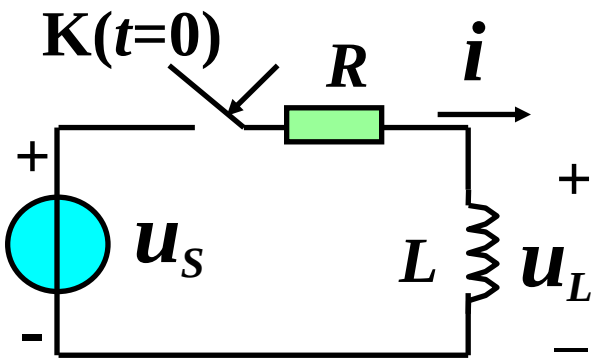






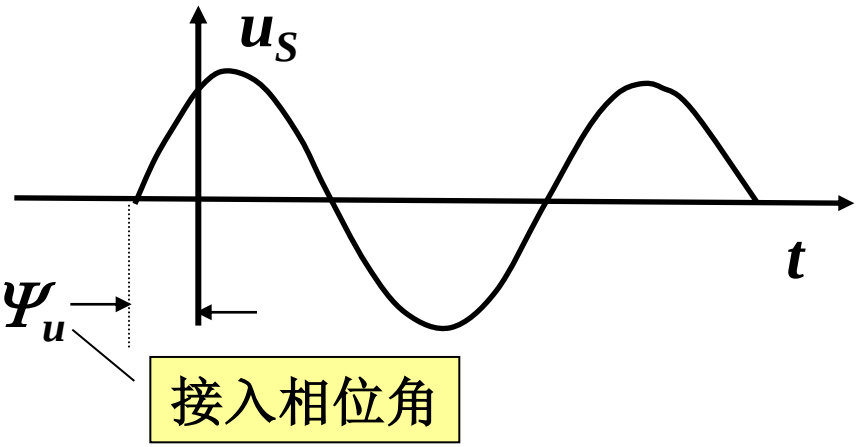
应用案例1：正弦电源激励下的一阶电路响应

例1



$i_L(0^-)=0$

$u_s = U_m \sin(\omega t + \psi_u)$　　$i\,(0^-)=0$



求：$i(t)$

用相量法计算稳态解　$I_m = \dfrac{U_m}{\sqrt{R^2+(\omega L)^2}}$　$\varphi = \text{arctg}\dfrac{\omega L}{R}$　$\tau = \dfrac{L}{R}$

$i\left(0^+\right)=0$

$$i' = I_m \sin(\omega t + \psi_u - \varphi)$$

解答为　$i = I_m \sin(\omega t + \psi_u - \varphi) + \left\{0 - I_m \sin(\psi_u - \varphi)\right\} e^{-\frac{t}{\tau}}$

解答为 $i = I_m \sin(\omega t + \psi_u - \varphi) - I_m \sin(\psi_u - \varphi) e^{-\frac{t}{\tau}}$

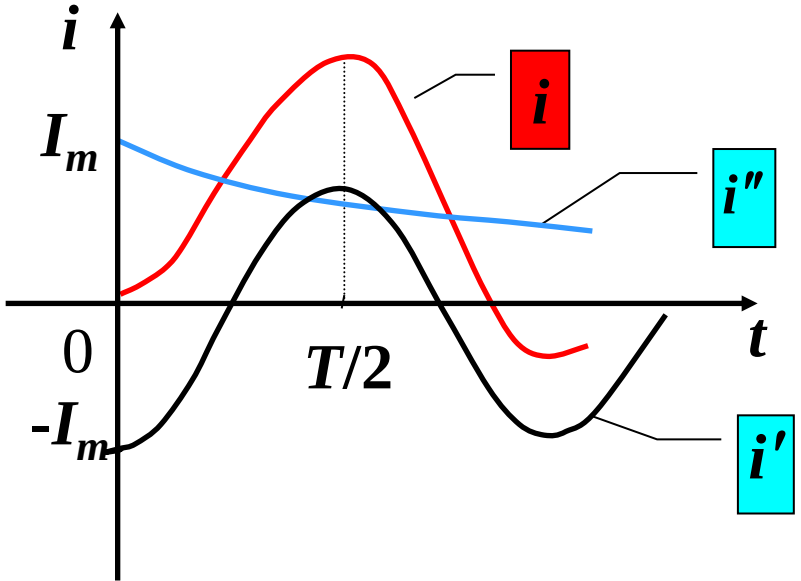
讨论几种情况：

1）合闸时 $\psi_u = \varphi$，则 $A = 0$，$i'' = 0$ 无暂态分量

电路直接进入稳态，不产生过渡过程。 $i = I_m \sin\omega t$

2） $\psi_u = \varphi \pm \pi/2$ 即 $\psi_u - \varphi = \pm\pi/2$ $i = I_m \sin(\omega t + \psi_u - \varphi) \mp I_m e^{-\frac{t}{\tau}}$

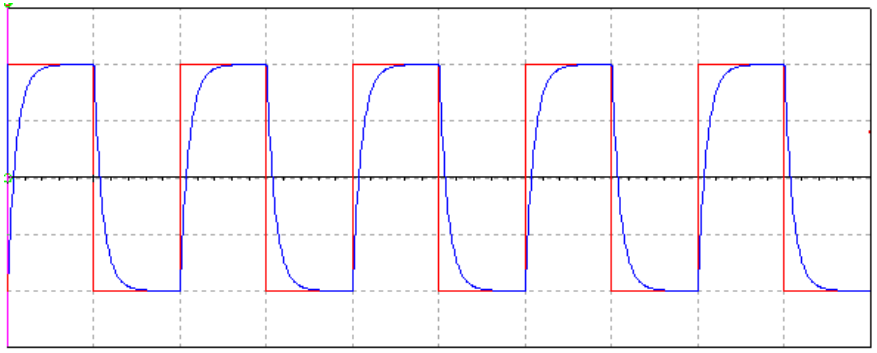
$\psi_u = \varphi - \pi/2$时波形为 $i = I_m \sin(\omega t - \pi/2) + I_m e^{-\frac{t}{\tau}}$

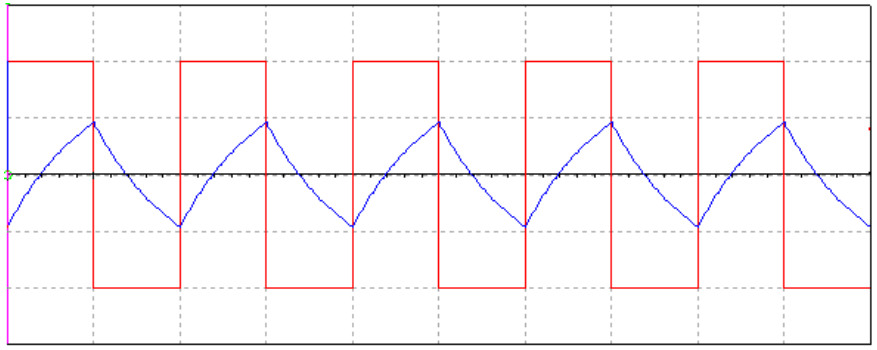


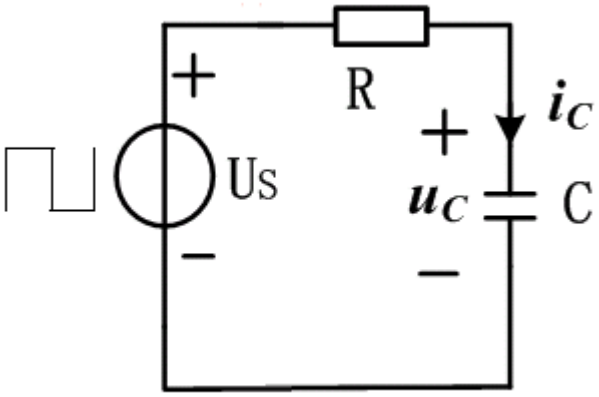
$i_{max} \le 2I_m$

最大电流出现在 $t = T/2$时刻。

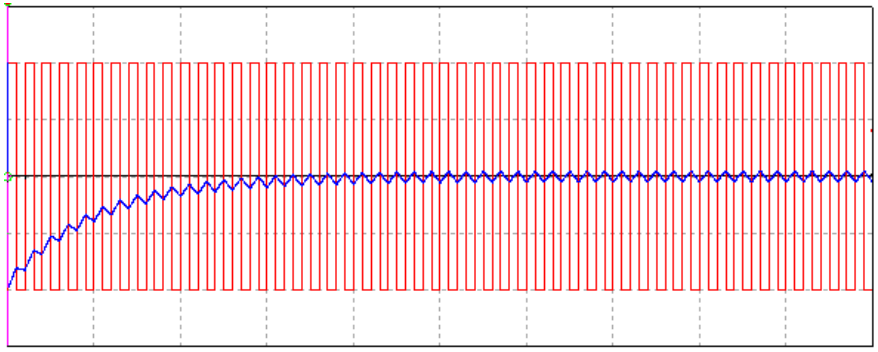
方波激励暂态响应示例

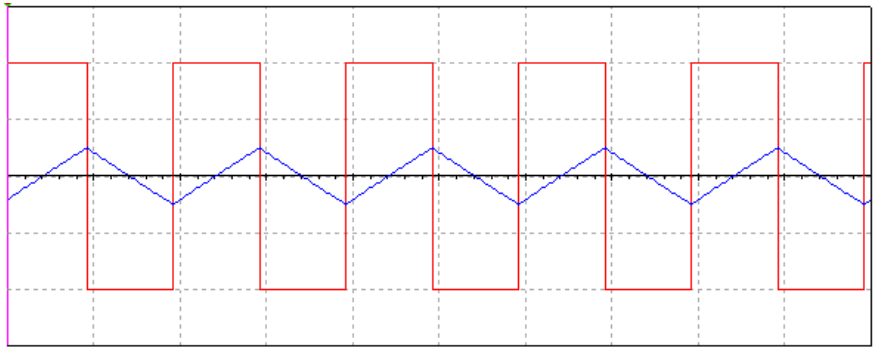




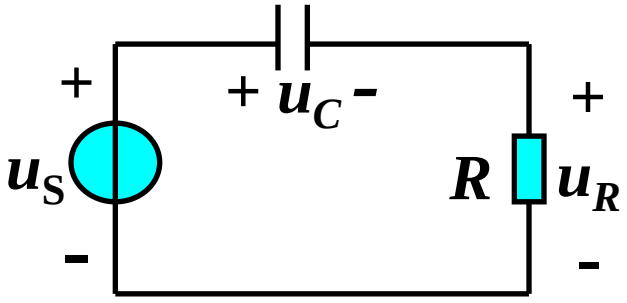


请选择合适的**R**、**C**和方波频率，实现下述形状的波形，并同时显示另一个元件的电压波形

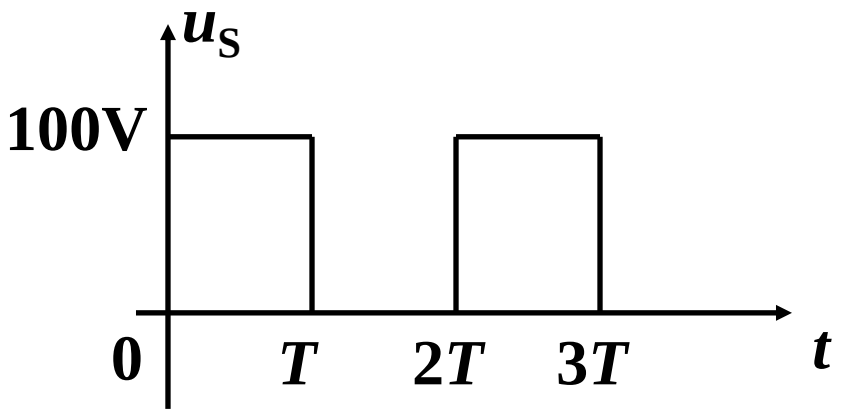




应用案例2： 脉冲序列作用下的*RC*电路



$u_C(0^-)=0$



（1） $T >> \tau$

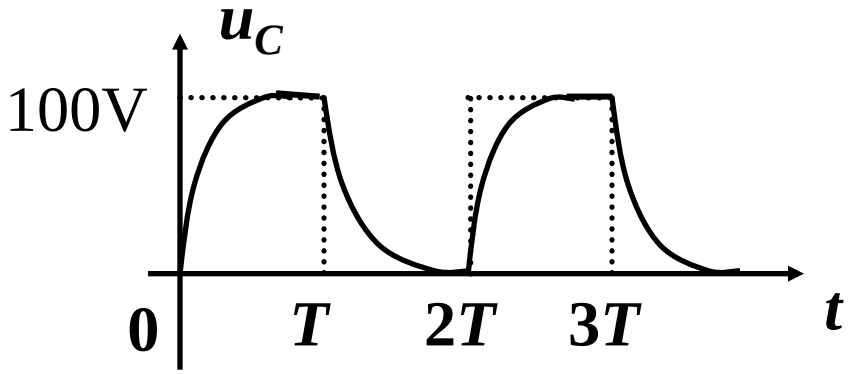
$0 < t < T$

$u_C(0^+)=0$

$u_C(\infty)=100\text{V}$

$\tau = RC$

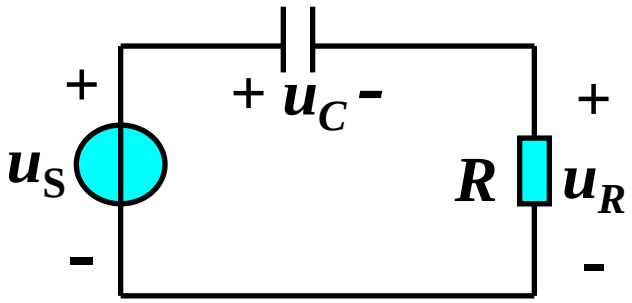
$$u_C = 100(1 - e^{-\frac{t}{RC}})\ \text{V}$$



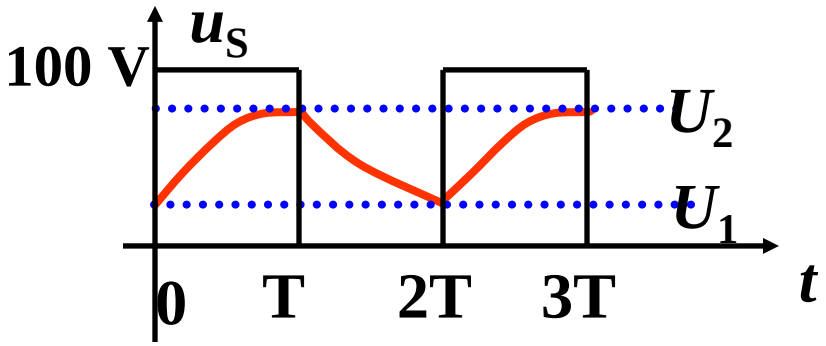
$T < t < 2T$ $\quad u_C(T^+)=100\text{V}$ $\quad u_C(\infty)=0$ $\quad \tau = RC$

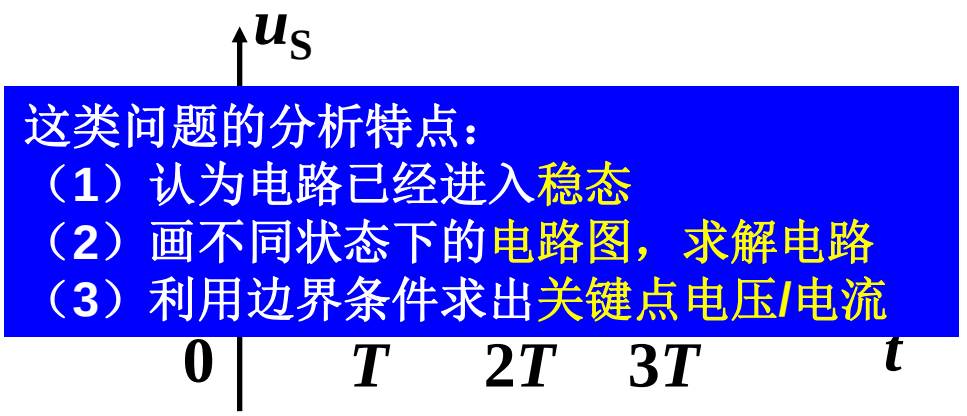
$$u_C = 100e^{-\frac{t-T}{RC}}\ \text{V}$$

（2） T 与 τ接近

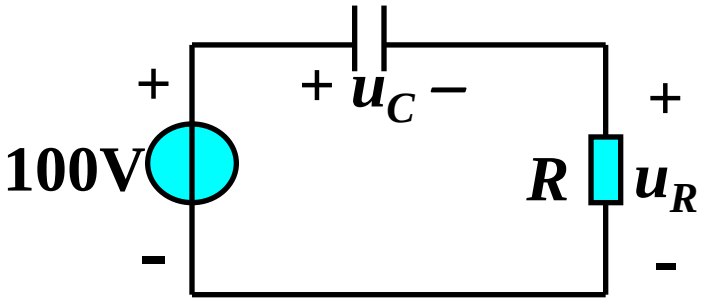


稳态解：　　$u_C(0^-)=0$





$0 < t < T$　　等效电路图

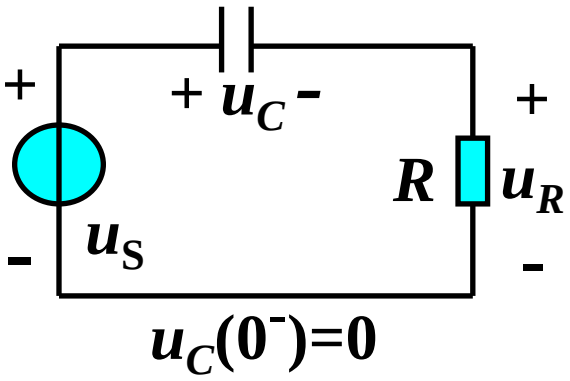


$$u_C(0^+)=U_1$$

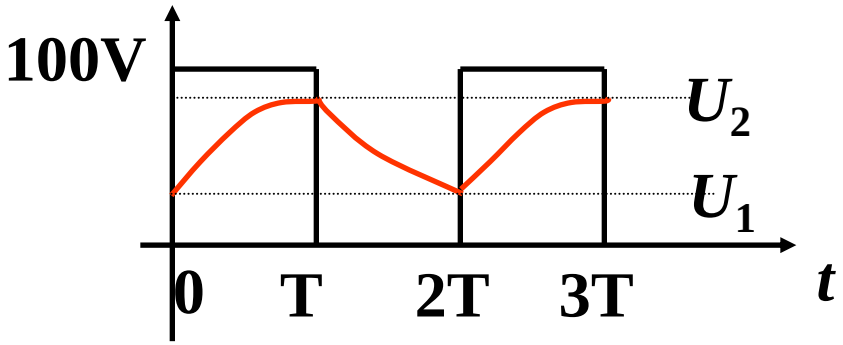
$$u_C(\infty)=100\ \text{V}$$

$$\tau = RC$$

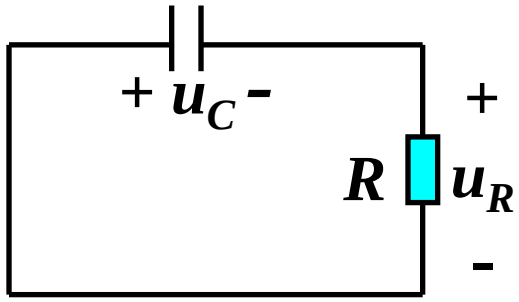
$$u_C = 100+(U_1-100)\mathrm{e}^{-\frac{t}{RC}}\ \text{V}$$



$u_C(0^-)=0$



$T < t < 2T$ 等效电路图

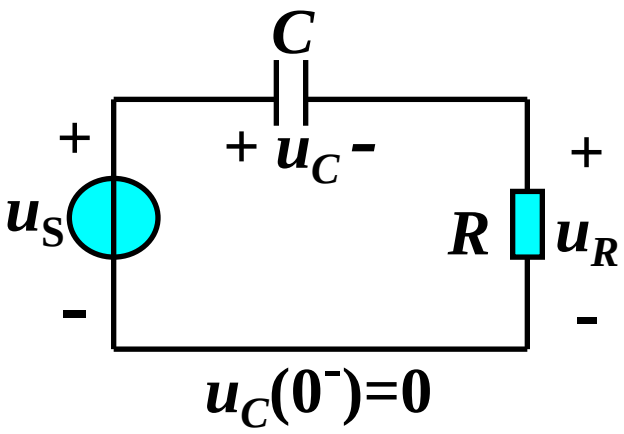


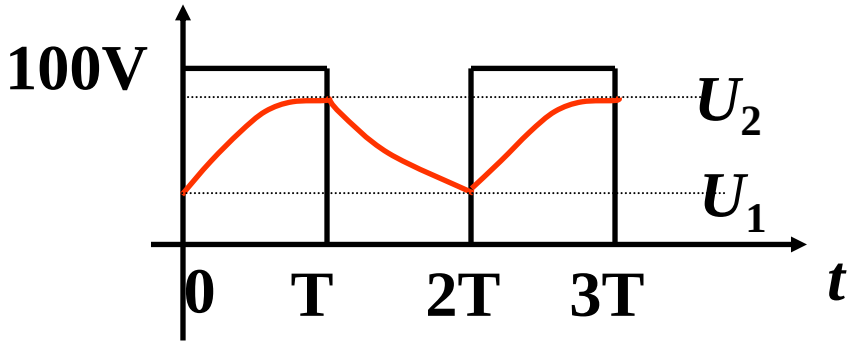
$$u_C(T^+)=U_2$$

$$u_C(\infty)=0$$

$$u_C = U_2 e^{-\frac{t-T}{RC}} \ \text{V}$$

$$\tau = RC$$





$0 < t < T$

$$u_C = 100 + (U_1 - 100)e^{-\frac{t}{RC}}$$

$T < t < 2T$

$$u_C = U_2 e^{-\frac{t-T}{RC}}$$

$t = T$

$$u_C = U_2 = 100 + (U_1 - 100)e^{-\frac{T}{RC}}$$

$t = 2T$

$$u_C = U_1 = U_2 e^{-\frac{2T-T}{RC}}$$

这类问题的分析特点：

（1）设电路已经进入稳态

（2）画电路图，求解电路

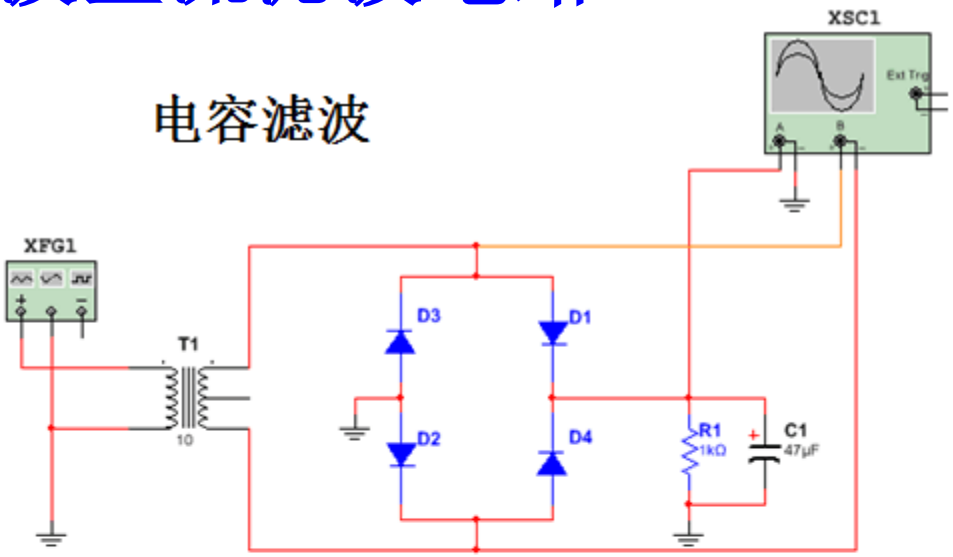
（3）利用边界条件求出

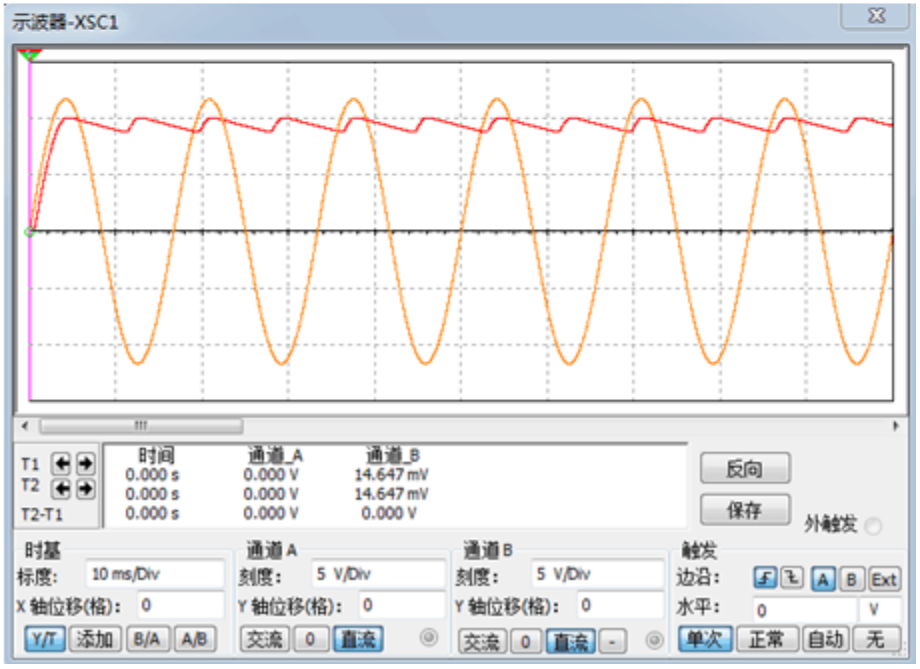
关键点电压/电流

$$U_1 = \frac{100e^{-\frac{T}{RC}}}{1+e^{-\frac{T}{RC}}}$$

$$U_2 = \frac{100}{1+e^{-\frac{T}{RC}}}$$

全波整流滤波电路


电容滤波
XSC1
XFG1
T1
10
D3
D1
D2
D4
R1
1kΩ
C1
47μF


示波器-XSC1
T1
T2
T2-T1
时间
通道_A
通道_B
0.000 s
0.000 V
14.647 mV
0.000 s
0.000 V
14.647 mV
0.000 s
0.000 V
0.000 V
反向
保存
外触发
时基
标度:
10 ms/Div
X 轴位移(格):
0
通道 A
刻度:
5 V/Div
Y 轴位移(格):
0
通道 B
刻度:
5 V/Div
Y 轴位移(格):
0
触发
边沿:
水平:
0
V
Y/T
添加
B/A
A/B
交流
0
直流
交流
0
直流
单次
正常
自动
无

第7章　动态电路的暂态分析

(dynamic circuit)　(transient analysis)

7.1 动态电路概述

7.2 电路的初始条件

7.3 一阶电路的暂态响应

7.4 一阶电路的阶跃和冲激响应

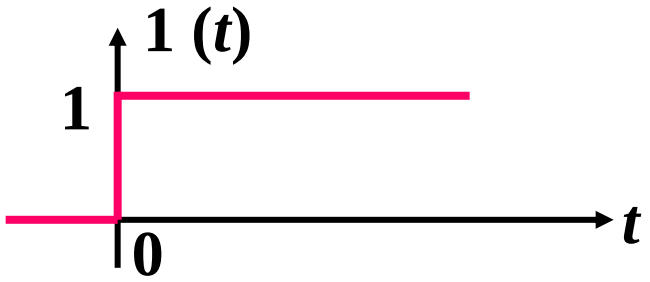
7.5 二阶电路的响应

7.6 高阶电路过渡过程的求解方法

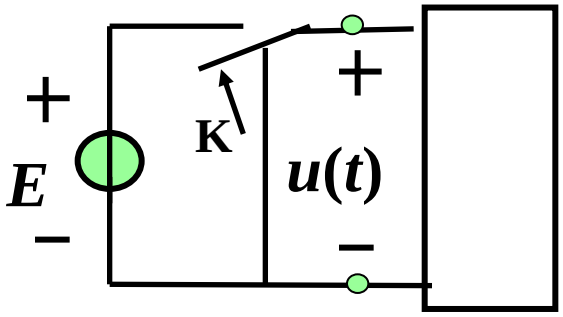
§7-5 一阶电路的阶跃响应和冲激响应

一　单位阶跃函数

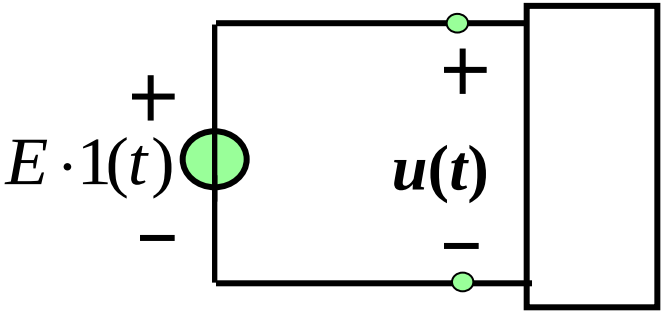
1. 定义　$1(t)=\begin{cases}0 & t<0\\ 1 & t>0\end{cases}$



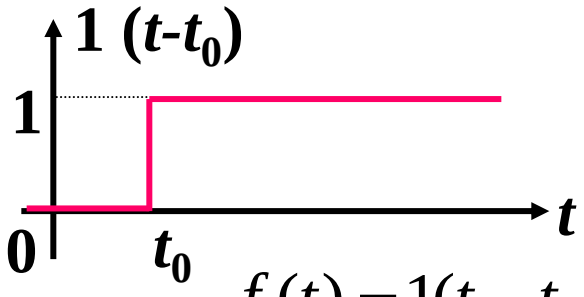
用 $1(t)$ 来描述开关的动作



$t=0$合闸　$u(t)=E\cdot 1(t)$

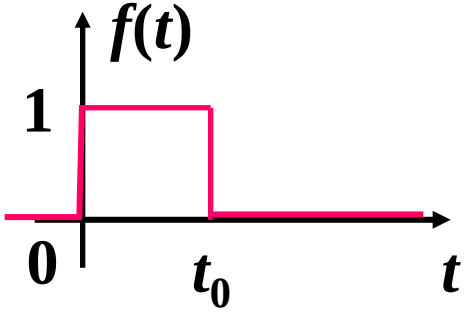


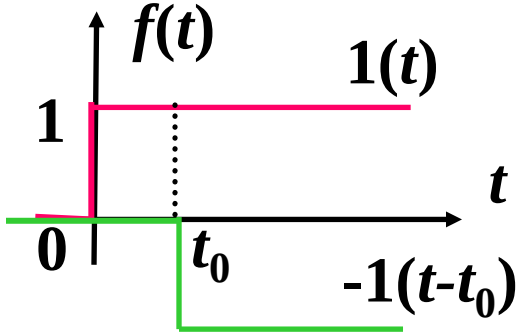
2. 单位阶跃函数的延迟



$$f(t)=1(t-t_0)$$

3. 由单位阶跃函数可组成复杂的信号





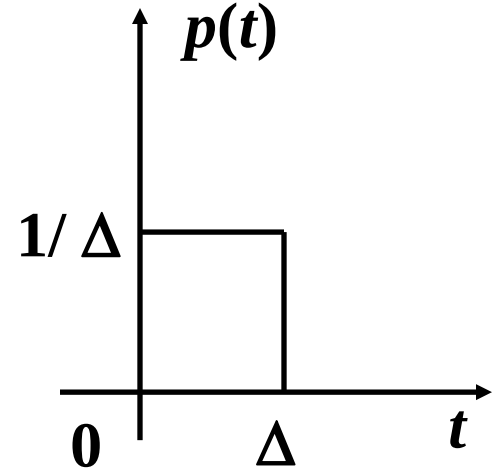
$$f(t)=1(t)-1(t-t_0)$$

二　单位冲激函数

1. 单位脉冲函数 $p(t)$

$$p(t)=\frac{1}{\Delta}[1(t)-1(t-\Delta)]$$

$$\int_{-\infty}^{\infty}p(t)\mathrm{d}t=1$$

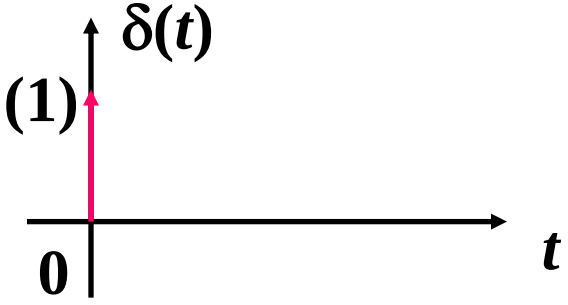


2. 单位冲激函数 $\delta(t)$

$$\Delta\to 0 \qquad \frac{1}{\Delta}\to\infty$$

$$\lim_{\Delta\to 0}p(t)=\delta(t)$$

定义

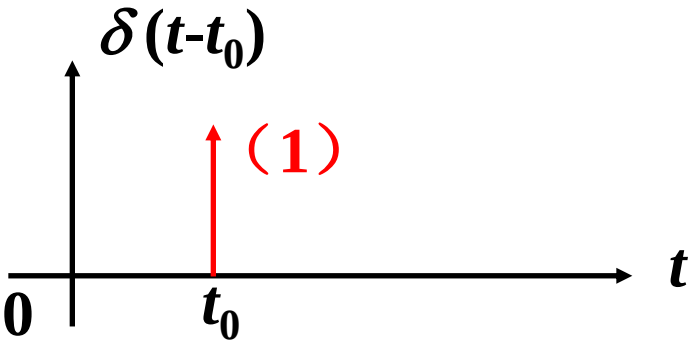


$$\delta(t)=\begin{cases}0 & (t<0)\\ 0 & (t>0)\end{cases}$$

$$\int_{-\infty}^{\infty}\delta(t)\mathrm{d}t=1$$

3. 单位冲激函数的延迟 $\delta(t-t_0)$

$$\begin{cases}\delta(t-t_0)=0 \quad (t\neq t_0)\\ \int_{-\infty}^{\infty}\delta(t-t_0)\mathrm{d}t=1\end{cases}$$

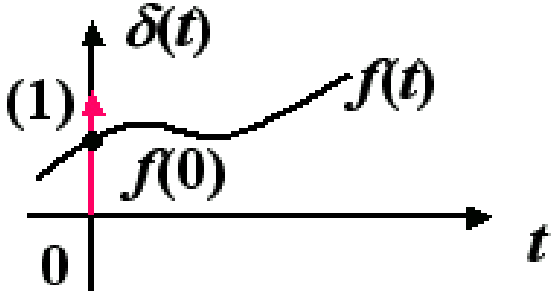


4. δ 函数的筛分性

$$\int_{-\infty}^{\infty} f(t)\delta(t)\mathrm{d}t = f(0)\int_{-\infty}^{\infty}\delta(t)\mathrm{d}t = f(0)$$

$f(0)\delta(t)$

同理有：$\int_{-\infty}^{\infty} f(t)\delta(t-t_0)\mathrm{d}t = f(t_0)$



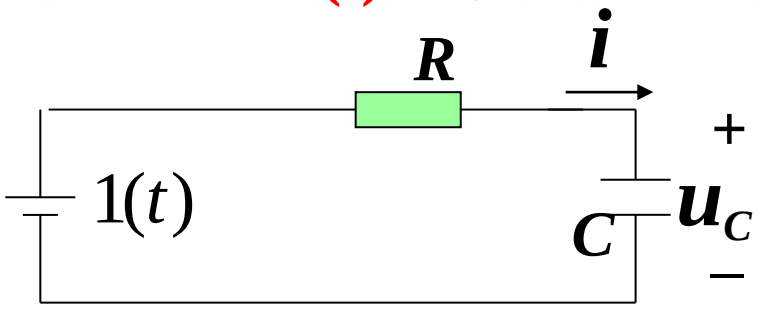
* $f(t)$在 t_0 处连续

5. δ(t)与1(t)的关系

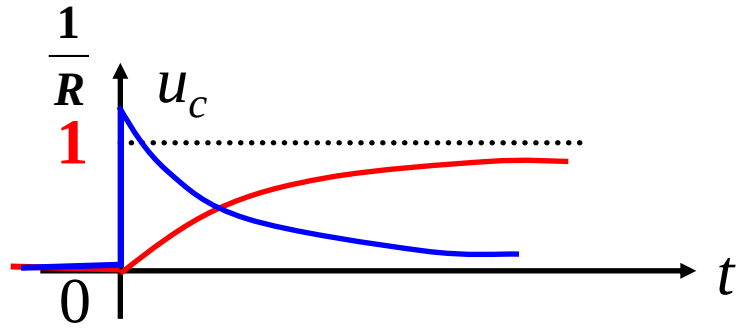
$$\delta(t)=\frac{d1(t)}{dt}$$

三　阶跃和冲激响应

1、阶跃响应s(t)：单位阶跃函数激励下电路中产生的零状态响应

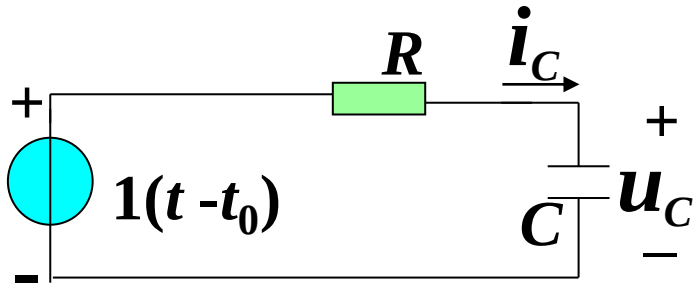


$u_C(0^-)=0$

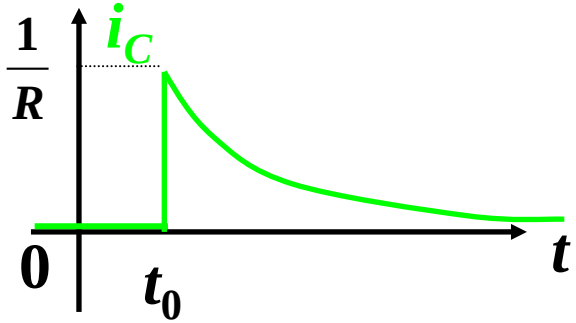


$$u_C(t)=(1-e^{-\frac{t}{RC}})\ 1(t)$$

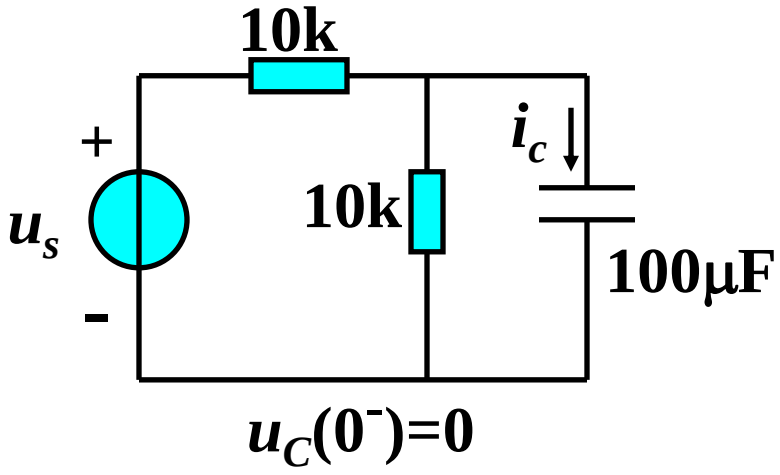
$$i(t)=\frac{1(t)-u_C}{R}=\frac{1}{R}e^{-\frac{t}{RC}}1(t)$$

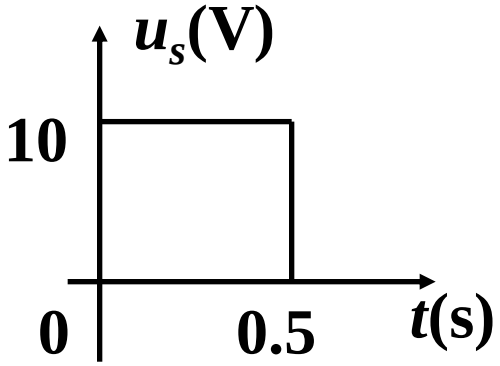


$$i_C=\frac{1}{R}e^{-\frac{t-t_0}{RC}}1(t-t_0)$$

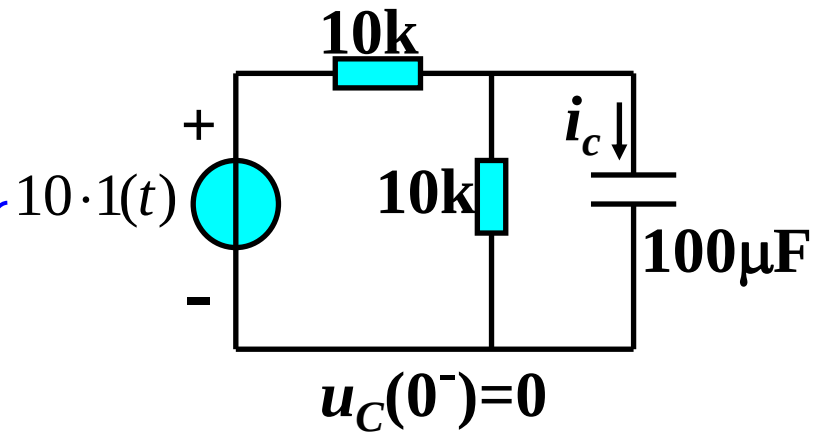


例 求图示电路中电流 $i_C(t)$

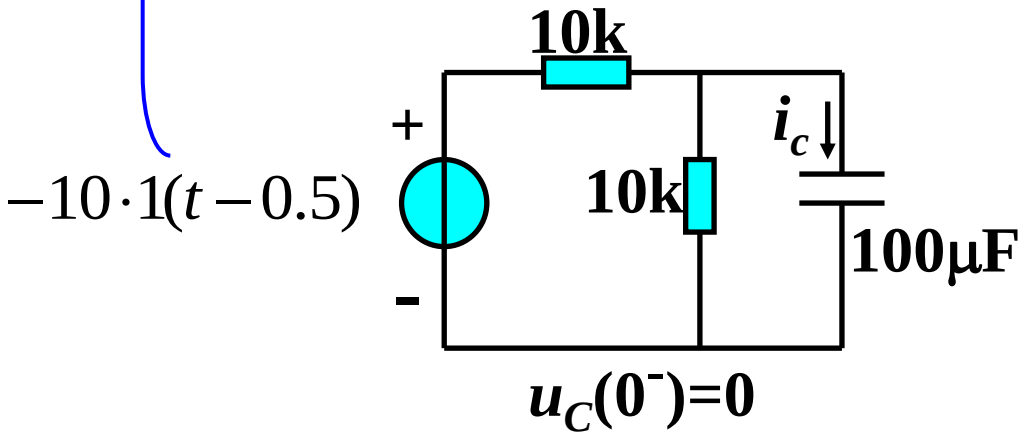




$$u_s = 10\cdot 1(t) - 10\cdot 1(t-0.5)$$



$$i_C = e^{-2t}1(t) \quad \text{mA}$$



$$i_C = -e^{-2(t-0.5)}1(t-0.5)\ \text{mA}$$

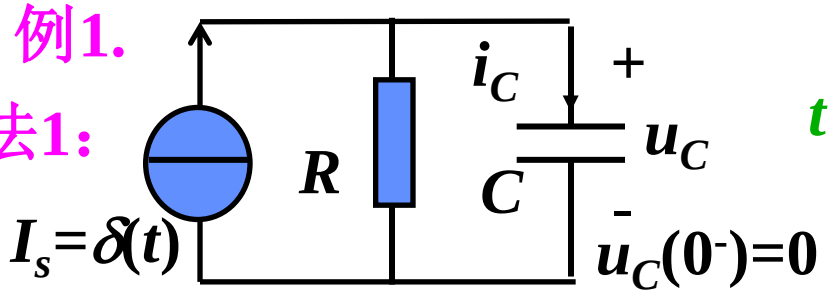
$$\therefore\ i_C = e^{-2t}1(t) - e^{-2(t-0.5)}1(t-0.5) \quad mA$$

2、冲激响应$h(t)$：单位冲激函数激励下电路中产生的零状态响应

1）分二个时间段来考虑冲激响应

例1.

解法1：



t { $0^- \longrightarrow 0^+$ 电容充电

$0^+ \longrightarrow \infty$ 零输入响应

(1). t 在 $0^- \longrightarrow 0^+$间 $\quad C\dfrac{du_c}{dt}+\dfrac{u_c}{R}=\delta(t)$

u_c 不可能是冲激函数，否则KCL不成立

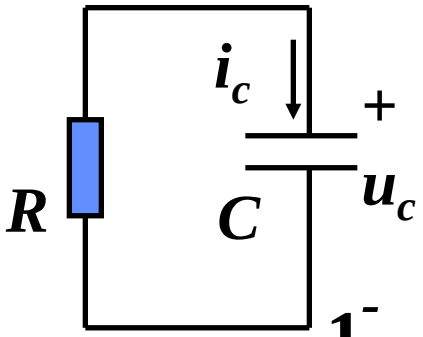
$$\int_{0^-}^{0^+} C\frac{du_c}{dt}dt+\int_{0^-}^{0^+}\frac{u_c}{R}dt=\int_{0^-}^{0^+}\delta(t)dt$$

$\int_{0^-}^{0^+}\frac{u_c}{R}dt=0$，$\int_{0^-}^{0^+}\delta(t)dt=1$

$$C[u_c(0^+)-u_c(0^-)]=1 \qquad u_c(0^+)=\frac{1}{C} \neq u_C(0^-)$$

电容中的冲激电流使电容电压发生跳变，即电源电流全部流过电容

(2). $t > 0^+$ 零输入响应 （RC放电）

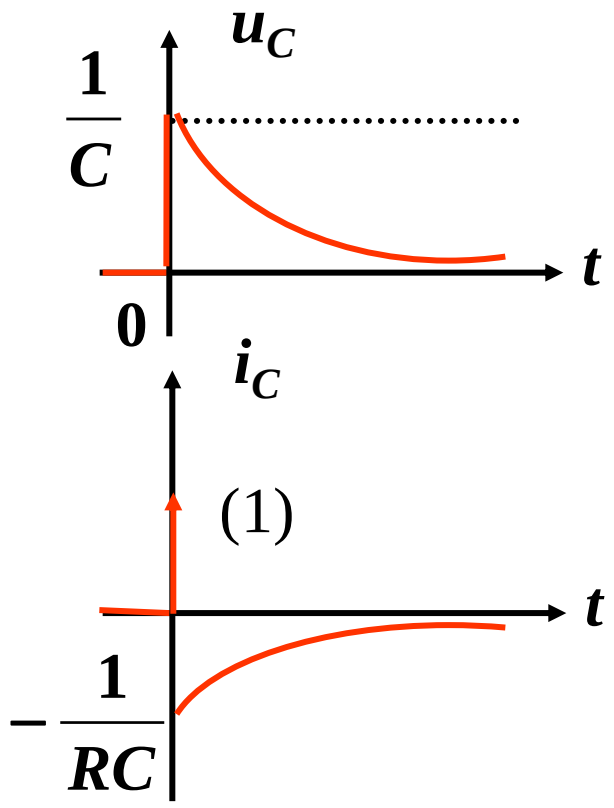


$$u_c(0^+) = \frac{1}{C}^-$$

$$u_c = \frac{1}{C} e^{-\frac{t}{RC}} \quad t \geq 0^+$$

$$i_c = -\frac{u_c}{R} = -\frac{1}{RC} e^{-\frac{t}{RC}} \quad t \geq 0^+$$

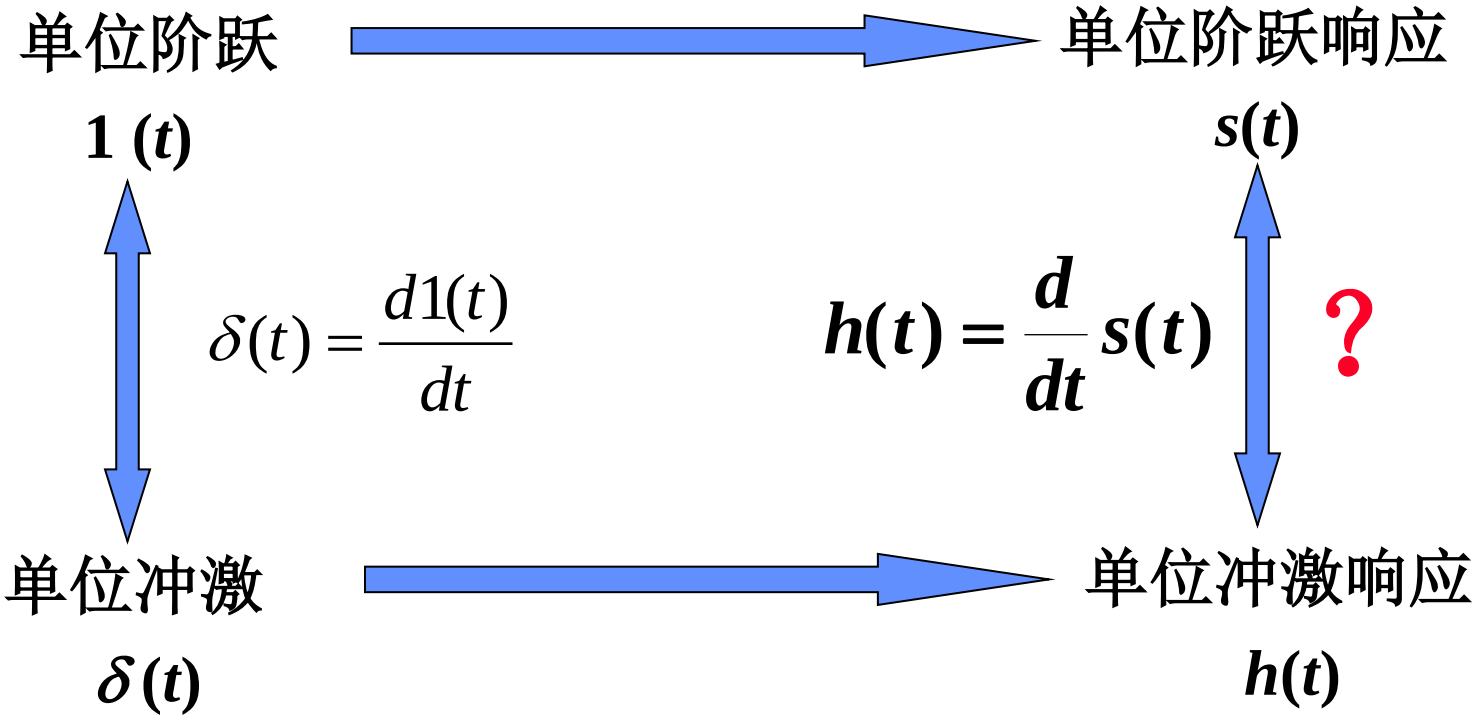
$$\begin{cases} u_c = \dfrac{1}{C} e^{-\frac{t}{RC}} 1(t) \\ i_c = \delta(t) - \dfrac{1}{RC} e^{-\frac{t}{RC}} 1(t) \end{cases}$$



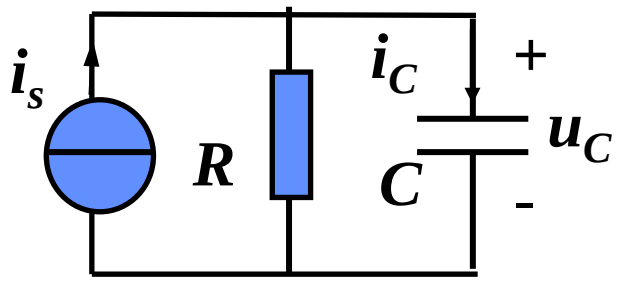
2）由单位阶跃响应求单位冲激响应



零状态相应与激励线性相关



例1解法2



已知：$u_c(0^-)=0$

求：$i_s(t)$为单位冲激时电路响应 $u_C(t)$和 $i_C(t)$

先求单位阶跃响应　令　$i_s(t)=1(t)$

$u_{Cs}(0^+)=0$　$u_{Cs}(\infty)=R$　$\tau=RC$　$i_{Cs}(0^+)=1$　$i_{Cs}(\infty)=0$

$$u_{Cs}(t)=R(1-e^{-\frac{t}{RC}})1(t)$$

$$i_{Cs}=e^{-\frac{t}{RC}}1(t)$$

全域表示

再求单位冲激响应　令　$i_s(t)=\delta(t)$

$$u_{Ch}=\frac{d}{dt}R(1-e^{-\frac{t}{RC}})1(t)=\underbrace{R(1-e^{-\frac{t}{RC}})\delta(t)}_{0}+\frac{1}{C}e^{-\frac{t}{RC}}1(t)$$

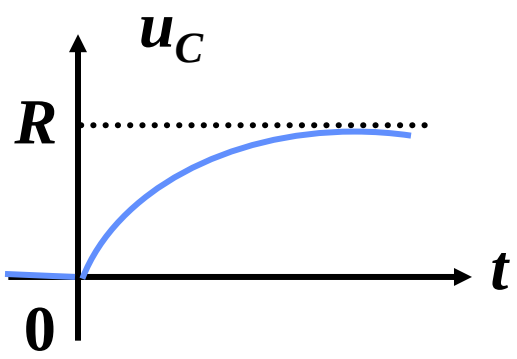
$$=\frac{1}{C}e^{-\frac{t}{RC}}1(t)$$

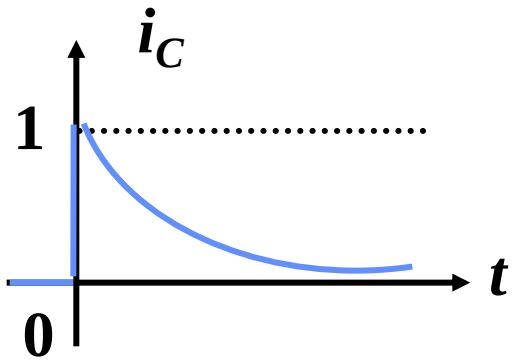
$f(t)\delta(t)=f(0)\delta(t)$

$$i_c = \frac{\mathrm{d}}{\mathrm{d}t}[e^{-\frac{t}{RC}}1(t)] = e^{-\frac{t}{RC}}\delta(t) - \frac{1}{RC}e^{-\frac{t}{RC}}1(t)$$

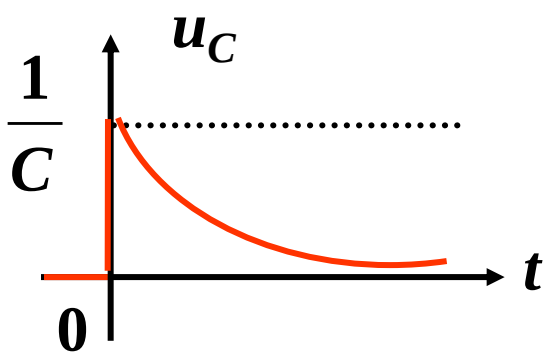
$$= \delta(t) - \frac{1}{RC}e^{-\frac{t}{RC}}1(t)$$

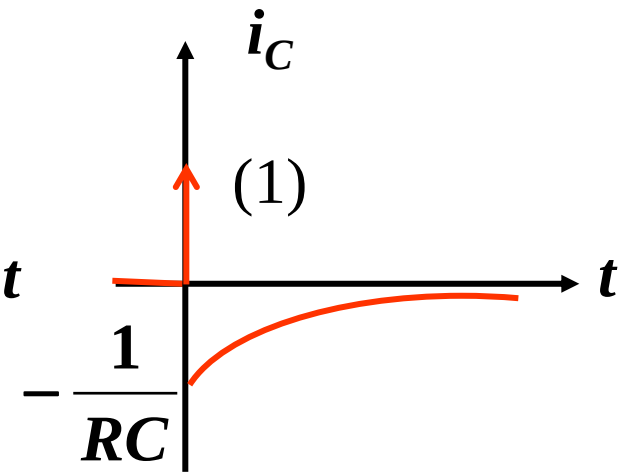
阶跃响应



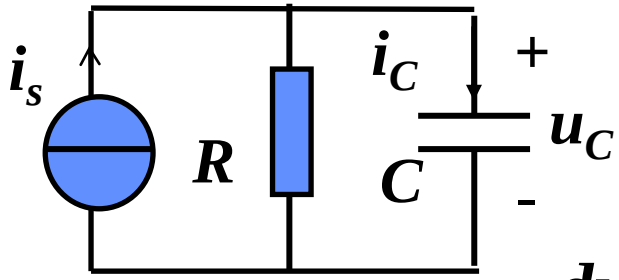


冲激响应





例1解法3



已知：$u_c(0^-)=0$

求：$i_s(t)$为单位冲激时电路响应 $u_C(t)$和 $i_C(t)$

$$C\frac{du_c}{dt}+\frac{u_c}{R}=\delta(t) \qquad u_c=\alpha\delta(t)+Ae^{-\frac{t}{RC}}1(t)$$

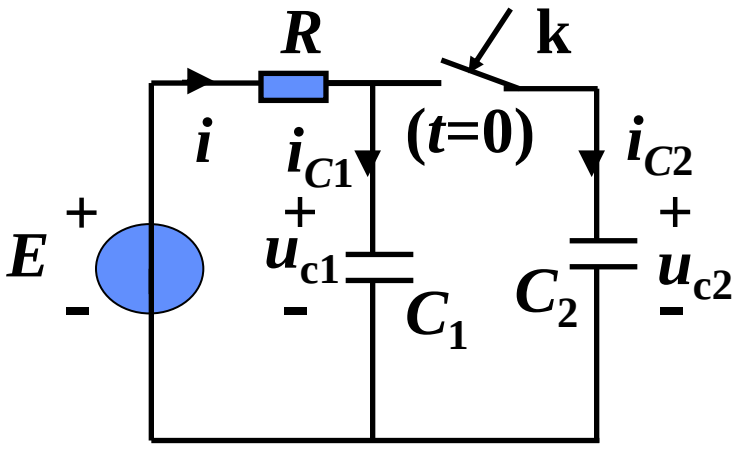
$$C\alpha\frac{d\delta(t)}{dt}+CAe^{-\frac{t}{RC}}\frac{d1(t)}{dt}-\frac{1}{RC}CAe^{-\frac{t}{RC}}1(t)$$

$$+\frac{1}{R}\left(\alpha\delta(t)+Ae^{-\frac{t}{RC}}1(t)\right)=\delta(t)$$

$$\alpha=0$$

$$CA+\frac{1}{R}\alpha=1$$

$$u_C=\frac{1}{C}e^{-\frac{t}{RC}}1(t)$$



例：奇异电路的暂态响应

已知：E=1V，R=1Ω，C_1=0.25F，C_2=0.5F，t = 0时合k。

求：u_{C1}，u_{C2}。

解　合k前　$u_{C1}(0^-)=E=1\text{V}$　　$u_{C2}(0^-)=0$

合k后　$u_{C1}(0^+)=u_{C2}(0^+)=u_C(0^+)$

电容电压初值一定会发生跳变。

$$\begin{cases} C_1u_{C1}(0^+)+C_2u_{C2}(0^+)=C_1u_{C1}(0^-)+C_2u_{C2}(0^-) \\ u_{C1}(0+)=u_{C2}(0+) \end{cases}$$

$u_C(\infty)$ =E= 1V

可解得 $u_C(0^+)=u_{C1}(0^+)=u_{C2}(0^+)=\frac{1}{3}\text{V}$

$\tau = R(C_1+C_2) = \frac{3}{4}\text{s}$

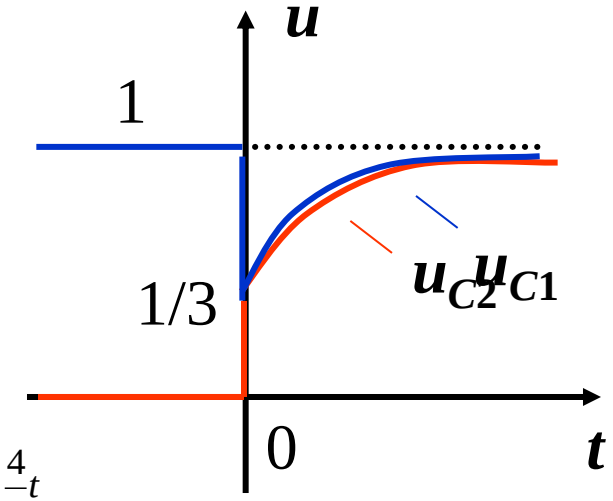
$$u_C(t)=1+(\frac{1}{3}-1)e^{-\frac{4}{3}t}=1-\frac{2}{3}e^{-\frac{4}{3}t}\quad t\geq 0^+$$

$$u_C(t) = 1 + (\frac{1}{3} - 1)e^{-\frac{4}{3}t} = 1 - \frac{2}{3}e^{-\frac{4}{3}t} \quad t \geq 0^+$$

$$t \leq 0^- \quad u_{C1}(0^-) = 1 \quad u_{C2}(0^-) = 0$$

$$u_{C1}(t) = 1(-t) + (1 - \frac{2}{3}e^{-\frac{4}{3}t})\,1(t)$$

$$u_{C2}(t) = (1 - \frac{2}{3}e^{-\frac{4}{3}t})\,1(t)$$



$$i_{C1} = C_1 \frac{\mathrm{d}u_1}{\mathrm{d}t}$$

$$= 0.25[-\delta(-t) + \frac{8}{9}e^{-\frac{4}{3}t}1(t) + (1 - \frac{2}{3}e^{-\frac{4}{3}t})\delta(t)]$$

$$= -\frac{1}{6}\delta(t) + \frac{2}{9}e^{-\frac{4}{3}t}1(t)$$

$$u_{C1}(t) = 1(-t) + (1 - \frac{2}{3}e^{-\frac{4}{3}t})1(t)$$

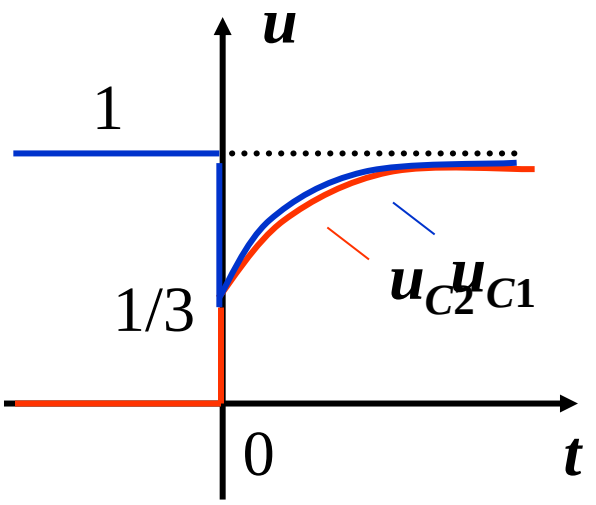
$$i_{C1}(t) = -\frac{1}{6}\delta(t) + \frac{2}{9}e^{-\frac{4}{3}t}1(t)$$

$$u_{C2}(t) = (1 - \frac{2}{3}e^{-\frac{4}{3}t})1(t)$$

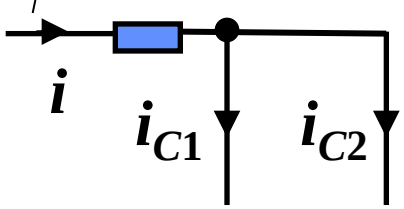
$$i_{C2}(t) = C_2\frac{\mathrm{d}u_2}{\mathrm{d}t}$$

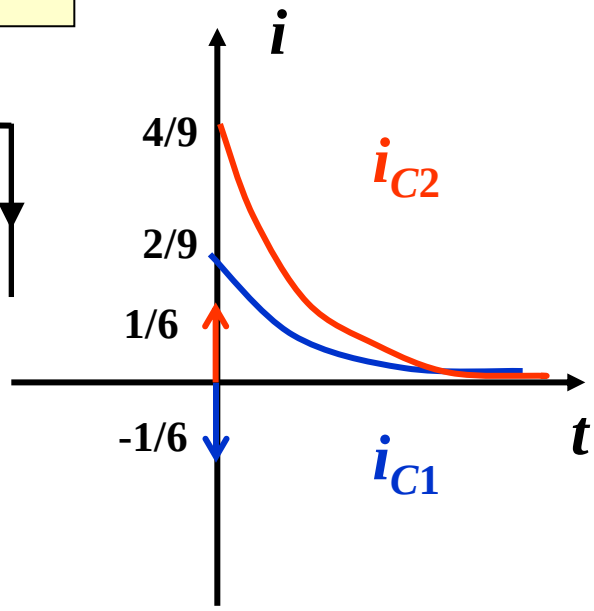
$$= 0.5[\frac{8}{9}e^{-\frac{4}{3}t}1(t) + (1 - \frac{2}{3}e^{-\frac{4}{3}t})\delta(t)]$$

$$= \frac{1}{6}\delta(t) + \frac{4}{9}e^{-\frac{4}{3}t}1(t)$$



i 无冲激





根据物理概念求电容电流

$0^- \rightarrow 0^+$

$$\Delta u_{C1} = \frac{1}{3} - 1 = -\frac{2}{3}$$

转移的电荷 $\Delta q_1 = 0.25 \times (-2/3) = -1/6$

$$\Delta u_{C2} = 1/3 - 0 = 1/3$$

$-1/6\ \delta(t)$ 冲激电流

转移的电荷 $\Delta q_2 = 0.5 \times 1/3 = 1/6$

$1/6\ \delta(t)$ 冲激电流

$t > 0^+$

$$i_{C1} = 0.25\frac{\mathrm{d}(1-\frac{2}{3}e^{-\frac{4}{3}t})}{\mathrm{d}t} = \frac{2}{9}e^{-\frac{4}{3}t}$$

$$i_{C2} = 0.5\frac{\mathrm{d}(1-\frac{2}{3}e^{-\frac{4}{3}t})}{\mathrm{d}t} = \frac{4}{9}e^{-\frac{4}{3}t}$$

$$\begin{cases} i_{C1} = -\frac{1}{6}\delta(t) + \frac{2}{9}e^{-\frac{4}{3}t}1(t) \\ i_{C2} = \frac{1}{6}\delta(t) + \frac{4}{9}e^{-\frac{4}{3}t}1(t) \end{cases}$$

作业

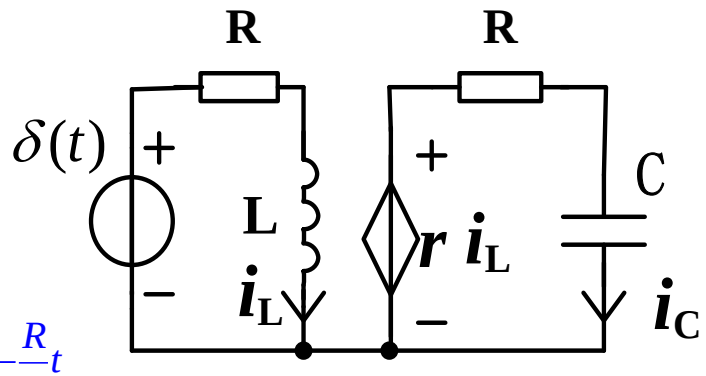
- 初值：7-2，3，4*，5*
- 一阶电路：7-6，7，12，13，14，15，
- 16（冲激）
- 二阶：7.18，20，22，24

讨论：7-17，27，28

含半导体器件：7-29-36，29，31，33，35，36

例题9: 如图电路, $u_C(0^-)=0, i_L(0^-)=0$

试求 $i_C(t)=0.$



解:

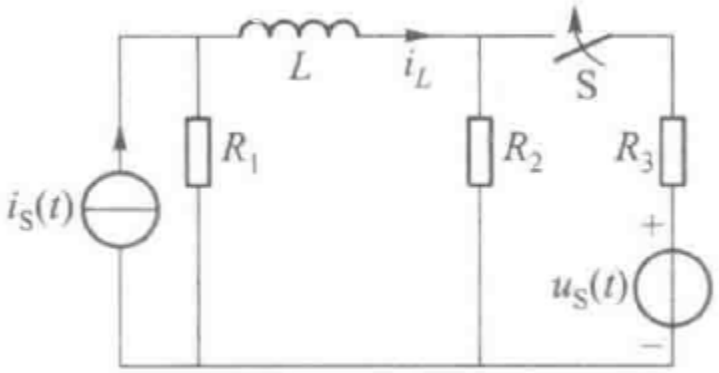
$\delta(t)$ $i_L(0^+)=\dfrac{1}{L}\neq i_L(0^-)$ $i_L(t)=i_L(0^+)e^{-\frac{R}{L}t}$

$$RC\frac{du_C}{dt}+u_C=\frac{\gamma}{L}e^{-\frac{R}{L}t}$$ 设特解为 $u_{C1}=Ae^{-\frac{R}{L}t}$

$1(t)$ $i_L(t)=i_{Lp}(t)+\left\{i_L(0^+)-i_{Lp}(0^+)\right\}e^{-\frac{R}{L}t}=\dfrac{1}{R}(1-e^{-\frac{R}{L}t})$

$$RC\frac{du_C}{dt}+u_C=\gamma\frac{1}{R}(1-e^{-\frac{R}{L}t})$$ 设特解为 $u_{C1}=Ae^{-\frac{R}{L}t}+B$

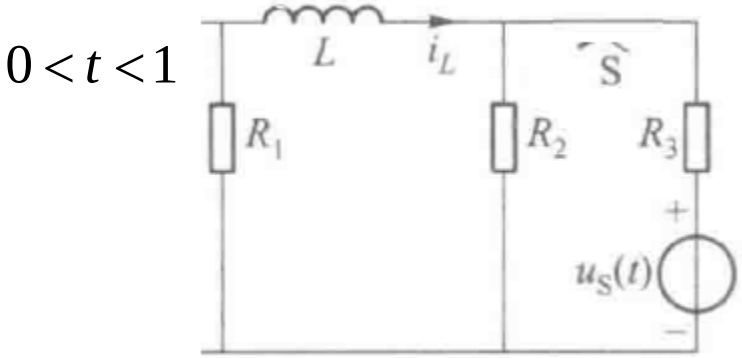
7.16 题图 7.16 所示电路中，开关 S 原已打开，已知 $L=1\ \mathrm{H}$，$R_1=R_2=R_3=1\ \Omega$，$i_L(0_-)=0$，$i_S(t)=\delta(t)\cdot A$，开关 $t=1\ \mathrm{s}$ 时闭合，试用经典法求 $u_S(t)=-1(t-1)\cdot \mathrm{V}$ 时的响应 $i_L(t)$。



$0<t<1$

$$i_{Ls}=i_s\frac{R_1}{R_1+R_2}\left(1-e^{-\frac{(R_1+R_2)t}{L}}\right)\varepsilon(t)$$

$$i_{Lh}=\frac{di_{Ls}}{dt}=i_s\frac{R_1}{L}e^{-\frac{(R_1+R_2)t}{L}}\varepsilon(t)$$



$$i_L=i_{Lh}+i_{Lt>1}$$

$$i_{Lt>1}=\frac{u_s}{1.5}\times 0.5+\left(\frac{i_sR_1}{L}-\frac{u_s}{1.5}\times 0.5\right)e^{-\frac{(R_3//R_2+R_1)(t-1)}{L}}\varepsilon(t-1)$$

$$7-16\quad i=\left[\frac{1}{3}+\left(e^{-2}-\frac{1}{3}\right)e^{-1.5(t-1)}\right]\cdot 1(t-1)+e^{-2t}\cdot\left[1(t)-1(t-1)\right]$$